https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	x is the independent variable and *V* is the dependent variable **or** vary *x* and measure *V*	**1**
	keep current (in the coil P) constant	**1**
	Methods of data collection	
	labelled diagram showing both coils supported	**1**
	two correct circuit diagrams for coil P and coil Q: power supply connected to one coil and voltmeter/c.r.o. connected to other coil	**1**
	method to determine *x*, e.g. use a ruler or drawn labelled horizontal ruler adjacent to coils with *x* indicated	**1**
	method to measure *x* from centre of coil P to centre of coil Q, e.g. measure width of (each) coil and divide by 2 and add to separation of coils	**1**
	Method of analysis	
	plots a graph of ln *V* against *x* [or log *V* against *x* etc.]	**1**
	relationship valid if a straight line produced	**1**
	k = –gradient	**1**

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Question	Answer	Marks
	Additional detail including safety considerations	**Max. 6**
	D1 do not touch hot coil/use gloves to position hot coil/heat-proof gloves to position coil	
	D2 use large current/number of turns/iron core (to produce large magnetic field/induced e.m.f.)	
	D3 use high frequency (to produce larger induced e.m.f.)	
	D4 use an a.c. power supply or signal generator (connected to coil P)	
	D5 keep the number of turns (on each coil) constant/frequency constant	
	D6 method described to check that current is constant, e.g. use an ammeter and variable resistor/variable power supply	
	D7 repeat measurements of x for different parts of the coil and average	
	D8 method to position ruler horizontally to measure x described e.g. use a spirit level or same height from bench at both ends	
	D9 method to keep coils parallel/co-axial e.g. adjust coil Q until maximum reading or use set square to ensure that coils are at right angles to the axis	
	D10 $\ln V = -kx + \ln V_0$	

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